Lab 7: Null Space and Inverses Solutions

EECS 245, Fall 2026 at the University of Michigan

due by the end of your lab section on Wednesday, October 21st, 2026

Name: _____

uniqname: _____

Each lab worksheet will contain several activities, some of which will involve writing code and others that will involve writing math on paper. To receive credit for a lab, you must complete all activities and show your lab TA by the end of the lab section.

While you must get checked off by your lab TA **individually**, we encourage you to form groups with 1-2 other students to complete the activities together.

Recap: Rank, Column Space, Null Space, and Inverses

- For an $n \times d$ matrix A, $\text{colsp}(A) \subseteq \mathbb{R}^n$ has dimension $\text{rank}(A)$.
- $\text{nullsp}(A) = \{\vec{x} \in \mathbb{R}^d : A\vec{x} = \vec{0}\}$, and $\dim(\text{nullsp}(A)) = d - \text{rank}(A)$.
- $\text{rank}(A^T) = \text{rank}(A)$.
- A square matrix A is invertible if and only if it has full rank. Its inverse satisfies $AA^{-1} = A^{-1}A = I$.

Activity 1: PrairieLearn Practice Problems

We're testing out a new website for practicing linear algebra problems: PrairieLearn.

Click **this link** to access the relevant problems for this activity. It consists of 6 problems, each worth 1 point. The numbers in the problems are randomized; everyone will receive slightly different problems.

To get credit for Activity 1, you must **eventually** correctly answer all 6 problems, earning a score of 6/6. If you answer a problem incorrectly, click "New Variant" to generate a new version and then try again. There is no penalty for answering a problem incorrectly, as long as you eventually get it correct.

To be clear, you don't need to include anything in your PDF for Activity 1; we will manually verify that you've finished all 6 problems.

If you have trouble accessing PrairieLearn, message Suraj on Slack.

Activity 2: Null Space of a Matrix with Linearly Independent Columns

Let $A = \begin{bmatrix} 3 & 0 \\ 0 & 4 \\ 1 & 0 \end{bmatrix}$. What is nullsp$(A)$?

Solution: The columns of A are linearly independent, so the only linear combination that gives $\vec{0}$ has both coefficients zero. Equivalently,

$$A \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3x_1 \\ 4x_2 \\ x_1 \end{bmatrix} = \vec{0} \quad \Longrightarrow \quad x_1 = x_2 = 0$$

Thus $\boxed{\text{nullsp}(A) = \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\}}$.

Activity 3: Null Space from Column Relationships

Let $X = \begin{bmatrix} 1 & 2 & -1 & 3 & 4 & 4 \\ 2 & 5 & -2 & 7 & 11 & 10 \\ 4 & 8 & -4 & 12 & 16 & 16 \end{bmatrix}$.

This is the matrix from Lab 6. Recall that its first two columns form a basis for its column space and its rank is 2.

a) Fill in the blanks: nullsp(X) is a ____-dimensional subspace of ____.

Solution:
From Chapter 5.4, the **rank-nullity theorem** states that for any $n \times d$ matrix X,

$$\text{rank}(X) + \dim(\text{nullsp}(X)) = \underbrace{d}_{\text{\# columns in } X}$$

The rank of X was found to be 2, and X has 6 columns, so

$$2 + \dim(\text{nullsp}(X)) = 6$$

$$\dim(\text{nullsp}(X)) = 6 - 2 = 4$$

By definition, nullsp(X) is the set of all vectors $\vec{v} \in \mathbb{R}^6$ such that $X\vec{v} = \vec{0}$. Therefore, it is a subspace of $\mathbb{R}^6$.

$$\boxed{\text{nullsp}(X) \text{ is a 4-dimensional subspace of } \mathbb{R}^6}$$

b) Find a basis for nullsp(X). *Hint: You should be able to answer this without solving equations.*

Solution: Let $\vec{v}_j$ be column j of X. The relationships

$$\vec{v}_1 + \vec{v}_3 = \vec{0}, \quad \vec{v}_1 + \vec{v}_2 - \vec{v}_4 = \vec{0}, \quad -2\vec{v}_1 + 3\vec{v}_2 - \vec{v}_5 = \vec{0}, \quad 2\vec{v}_2 - \vec{v}_6 = \vec{0}$$

give the following four vectors in the null space:

$$\boxed{\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 3 \\ 0 \\ 0 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 0 \\ 0 \\ 0 \\ -1 \end{bmatrix} \right\}}$$

They are linearly independent: looking at coordinates 3, 4, 5, and 6 in a linear combination equal to zero forces all four coefficients to be zero. Since the null space has dimension 4, these four independent vectors form a basis.

Activity 4: The Systems of Equations View

Let A be an $n \times d$ matrix of rank r, and suppose there exist vectors $\vec{b} \in \mathbb{R}^n$ such that

$$A\vec{x} = \vec{b}$$

does not have a solution (meaning no $\vec{x}$ makes $A\vec{x} = \vec{b}$).

a) What are all inequalities ($<$ or $\leq$) that must be true between n, d, and r?

Solution:
If $A\vec{x} = \vec{b}$ has no solution for some $\vec{b}$, then $\vec{b}$ does not lie in colsp(A). Thus, the column space of A is a proper subspace of $\mathbb{R}^n$, meaning its dimension (the rank) is strictly less than n:

$$r < n$$

By general rank properties, rank cannot exceed either the number of rows or columns:

$$r \leq n, \quad r \leq d$$

Combining these gives:

$$\boxed{r < n \quad \text{and} \quad r \leq d}$$

b) How do you know that $A^T\vec{y} = \vec{0}$ has solutions other than $\vec{y} = \vec{0}$?

Solution:
By the rank-nullity theorem,

$$\text{rank}(A^T) + \dim(\text{nullsp}(A^T)) = \underbrace{n}_{\#\text{ columns in } A^T}$$

and since $\text{rank}(A^T) = r < n$, it follows that $\dim(\text{nullsp}(A^T)) = n - r > 0$. Therefore, nullsp$(A^T)$ contains non-zero vectors, so there must exist some $\vec{y} \neq \vec{0}$ such that $A^T\vec{y} = \vec{0}$.

Activity 5: Symbolic Inverses

Given that A is an invertible $n \times n$ matrix that satisfies $A^4 - 3A^2 + 2A - 4I = 0$, find an expression for A^{-1} in terms of A.

Solution: The goal is to find another matrix B such that $AB = BA = I$. We can do this by isolating the identity matrix, I, and then trying to write the other side of the equation as A times some other matrix.

$$\begin{aligned} A^4 - 3A^2 + 2A - 4I &= 0 \\ A^4 - 3A^2 + 2A &= 4I \\ A(A^3 - 3A + 2I) &= 4I \\ A\left(\frac{1}{4}(A^3 - 3A + 2I)\right) &= I \end{aligned}$$

So, $\boxed{A^{-1} = \frac{1}{4}(A^3 - 3A + 2I)}$.

Because A is invertible, multiplying $A(A^3 - 3A + 2I) = 4I$ on the left by A^{-1} proves the formula. Also, any polynomial in A commutes with A, so the displayed candidate satisfies both $AB = I$ and $BA = I$.

Activity 6: Basics of Invertibility

Suppose A is an $n \times n$ matrix. State as many of the equivalent conditions for invertibility as you can.

Solution: If A is an $n \times n$ matrix, then A is invertible if and only if:

- rank$(A) = n$
- A's columns are linearly independent (and hence colsp$(A) = \mathbb{R}^n$)
- A's rows are linearly independent (and hence rowsp$(A) =$ colsp$(A^T) = \mathbb{R}^n$)
- A's null space is only the zero vector, i.e. nullsp$(A) = \{\vec{0}\}$
- det$(A) \neq 0$

Activity 7: Invertibility of X^TX

a) Let X be an $n \times d$ matrix. Fill in the blanks: X^TX is invertible if and only if X's columns are ____.

Solution: $X^T X$ is invertible if and only if X's columns are **linearly independent**. This is because rank(X) = rank$(X^T X)$, as we proved in **Chapter 5.4**, and a square matrix is invertible if and only if its rank is equal to its number of columns.

For rank$(X^T X) = d$, we need rank$(X) = d$, meaning X's columns must be linearly independent.

Activity 8: Doing Things Differently

Matrix multiplication does not always behave like multiplication of numbers. All matrices below are 2×2, and all vectors are in $\mathbb{R}^2$. Justify each answer. For a counterexample, give specific matrices or vectors and show the relevant products.

a) If $B\vec{v} = \vec{0}$, must $AB\vec{v} = \vec{0}$? If $AB\vec{v} = \vec{0}$, must $B\vec{v} = \vec{0}$? Address both directions.

Solution: Yes in the first direction: $AB\vec{v} = A(B\vec{v}) = A\vec{0} = \vec{0}$. The converse is false. Take

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad B = I, \quad \vec{v} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Then $B\vec{v} = \vec{v} \neq \vec{0}$, but $AB\vec{v} = A\vec{v} = \vec{0}$.

b) Can $AB = 0_{2\times 2}$ even when neither A nor B is the zero matrix? Give an example or explain why it is impossible.

Solution: Yes. For example,

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \qquad AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Both factors are nonzero.

c) Expand $(A + B)^2$ using distributivity, keeping the factors in their original order. Is it always equal to $A^2 + 2AB + B^2$? If not, give a counterexample.

Solution: Distributivity gives

$$(A + B)^2 = A^2 + AB + BA + B^2$$

This equals $A^2 + 2AB + B^2$ if and only if $AB = BA$, which need not hold. For example, take

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

Then $A^2 = A$, $B^2 = 0$, $AB = B$, and $BA = 0$. Hence

$$(A + B)^2 = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \neq \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} = A^2 + 2AB + B^2$$